# PingSheng Kao

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## SUMMARY

Research Engineer and ML Data Scientist with 6+ years of experience evaluating human-AI interaction and improving multimodal, agentic, and production ML systems. Builds human-calibrated evaluations, behavioral metrics, simulated users, failure-analysis tools, and model-development feedback loops grounded in product experience.

## EXPERIENCE

**ML Data Scientist - Apple** July 2020 - Present
Santa Clara, California

- Translated human-factors research into evaluation protocols and metrics for per-turn intent misalignment and false updates to an agent's inferred user state, estimating behavioral occurrence rates at scale.
- Built simulated-user and agent-as-judge workflows for long multimodal interactions, with targeted scenarios for visual-memory and sycophancy weaknesses; systematic failures changed subsequent development direction.
- Developed a reference-based, multi-stage LLM evaluator and expert/non-expert validation workflow used internally for reporting and failure discovery; reached 90% expert alignment on a 40-example gold set.
- Scaled LLM- and agent-as-judge evaluation on an existing internal batch platform, debugging throughput and rate limits, versioning evaluators, and surfacing results through dashboards.
- Evaluated factual consistency and tool-call behavior, and built an internal chat prototype using VLM-as-judge retrieval and Voxel51/FiftyOne image collages for natural-language investigation of vision-model failures.
- Personally fine-tuned vision models on VLM-discovered and relighting-derived failure slices, comparing targeted model repair with alternative slice-discovery approaches.

## RESEARCH PUBLICATION

**GH-ESD: Grounded Hypothesis-Driven Error Slice Discovery for Instance-Level Vision Tasks** ECCV 2026
Listed author (personal contribution: experiments) - Apple ML Research

- Listed author and experiment contributor on the ECCV 2026 publication GH-ESD, developing grounded, hypothesis-driven error-slice discovery for instance-level vision tasks.

## ADDITIONAL EXPERIENCE

**Algorithm and Architecture Intern - Deep Learning Compression - Ambarella** June 2019 - August 2019
Santa Clara, California

- Developed a saliency-based neural-network quantization heuristic that reduced an exponential search formulation to a linear-time method, and proposed a mixed-precision algorithm integrated into company software.

## EDUCATION

**M.S. in Computer Science and Engineering - University of Michigan, Ann Arbor** 2018 - 2020
**B.S. in Electronic Engineering - National Taiwan University** 2013 - 2017
GPA: 3.9/4.0 (B.S.)

## SKILLS

Programming & ML: Python, SQL, C++, JavaScript, Model Fine-Tuning, Experiment Design
Model Evaluation: Agent-as-Judge, LLM-as-Judge, Simulated Users, Human Evaluation, DSPy
Human-AI Interaction: Intent Alignment, User-State Metrics, Sycophancy, Tool-Call Evaluation, Human Factors
Analysis: Metric Design, Embeddings, Clustering, OOD Analysis, Voxel51/FiftyOne, Bayesian Analysis